List of Publications

Journal articles (peer reviewed)

Adrian-Kalchhauser, I., Bussmann-Charran, K., N'Guyen, A., Hirsch, P. E., **Wiegleb, J.**, Bergmann, L., & Burkhardt-Holm, P. (2026). Factors Supporting Translational, Participatory Ecology Projects—Observations from Ten Years of Collaborative Research on Invasive Fish. *Environmental Management*, *76*(8), 253. https://doi.org/10.1007/s00267-026-02461-w

Avramović, M., Rashidabadi, F., Leandersson, D., Hajiesmaeili, M., **Wiegleb, J.**, Piccolo, J. J., & Watz, J. (2026). Drift feeding by brown trout fry under downramping flow conditions: effects of substrate structure. *Ecohydrology & Hydrobiology*, *26*(3), 100755. https://doi.org/10.1016/j.ecohyd.2026.100755

Govindasamy, N., Rauter, G., Seidel, F., Burkhardt-Holm, P., Hirsch, P. E., & **Wiegleb, J**. (2024). Does swimming at the bottom serve as a hydraulic advantage for benthic fish *Neogobius melanostomus* Pallas (1814) in flowing water? *Biology open*, *13*(11), bio060533. https://doi.org/10.1242/bio.060533

Wiegleb, J., Hirsch, P. E., Seidel, F., Rauter, G. & Burkhardt-Holm, P. (2023). Round goby [*Neogobius melanostomus* (Pallas, 1814)], gudgeon (*Gobio gobio* L.) and bullhead (*Cottus gobio* L.) show distinct swimming patterns in a vertical slot fish pass, Frontiers in Environmental Science, 11, 1156248 https://doi.org/10.3389/fenvs.2023.1156248

Wiegleb, J., Hirsch, P. E., Seidel, F., Rauter, G. & Burkhardt-Holm, P. (2023). Impact of hydraulic forces on the passage of round goby *(Neogobius melanostomus)*, gudgeon *(Gobio gobio)* and bullhead *(Cottus gobio)* in a vertical slot fish pass, *Ecology of Freshwater Fish*, 32(2), 416-430 https://doi.org/10.1111/eff.12696

Wiegleb, J., Hirsch, P. E., Seidel, F., Rauter, G. & Burkhardt-Holm, P. (2022). Flow, force, behaviour: assessment of a prototype hydraulic barrier for invasive fish, *Hydrobiologia,* 849, 1001-1019 https://doi.org/10.1007/s10750-021-04762-z

Egger, B., **Wiegleb, J.**, Seidel, F., Burkhardt-Holm, P., & Emanuel Hirsch, P. (2021). Comparative swimming performance and behaviour of three benthic fish species: The invasive round goby (*Neogobius melanostomus*), the native bullhead (*Cottus gobio*), and the native gudgeon (*Gobio gobio*). *Ecology of Freshwater Fish*, 30(3), 391-405 . https://doi.org/10.1111/eff.12592

Wiegleb, J., Hirsch, P. E., Egger, B., Seidel, F., & Burkhardt-Holm, P. (2020). Flow field-induced drag forces and swimming behavior of three benthic fish species. *Limnologica*, 84, 125812. https://doi.org/10.1016/j.limno.2020.125812

Lutz, E., Hirsch, P. E., Bussmann, K., **Wiegleb, J.**, Jermann, H.-P., Muller, R., … Adrian-Kalchhauser, I. (2020). Predation on native fish eggs by invasive round goby revealed by species-specific gut content DNA analyses. *Aquatic Conservation: Marine and Freshwater Ecosystems,* 30(8), 1566-1577. https://doi.org/10.1002/aqc.3409

Wiegleb, J., Kotterba, P., Hammer, C., & Oesterwind, D. (2018). Predation of the round goby (*Neogobius melanostomus* Pallas, 1814) on Atlantic herring eggs in the Western Baltic Sea. *Marine Biology Research*, *14*(9–10), 989-1003. https://doi.org/10.1080/17451000.2019.1577977

Manuscripts in preparation

Wiegleb, J., Vowles, A., Kurt, M., & Kemp, P. Hydrodynamic stimuli affect route preference of upstream navigating fish. In preparation

Durtsche, R. D., **Wiegleb, J.** Fish Rest: A retrofit solution for culverts providing resting stations for fish with low swimming performance. In preparation

Wiegleb, J., Silva, A. T., Bærum, K. M., Cretois, B., Georganos, S., Aldvén, D., Forseth, T., Calles, O. Where do eels decide? Spatially clustered commitment zones in downstream migration past a hydropower facility. In preparation

Patents/Intellectual Property

Durtsche, R., **Wiegleb, J.** (2026). Fish Rest. Swedish Intellectual Property Office (PRV), Application No. 2630382-6, Filed May 28, 2026. Status: Pending

Contributions to Conferences, Presentations, and Posters

Poster	Durtsche, R., **Wiegleb, J.** (2026) Take a Rest! Fish Rest provides resting stations for small-bodied fish in fast waters of culverts and fishways (Hydropower R&D days, Karlstad)
Presentation	Calles, O., **Wiegleb, J**. (2025) Improving downstream eel passage (FishPath workshop in Trondheim, Norway)
Presentation	**Wiegleb, J.** (2025) Improving eel downstream passage with machine learning (Hydropower R&D days, Uppsala)
Presentation	**Wiegleb, J.**, Vowles, A., Kurt, M., Kemp, P. (2024) Improving fish pass design via understanding fish swimming patterns in challenging flow (International Symposium on Ecohydraulics and Fish Passage, Québec, Canada).
Poster	**Wiegleb, J.** & Kemp, P. (2023) Lessons from a gummy fish: How do fish modulate their swimming trajectories when swimming through critical flow? (Fisheries Society of the Bristish Isles (FSBI) Symposium 2023 in Colchester, UK). http://dx.doi.org/10.13140/RG.2.2.24125.64480

Poster	**Wiegleb, J.**, Hirsch, P. E., Seidel, F., Rauter, G. & Burkhardt-Holm, P. (2022). Swimming competition! Who beats the flow in a hydraulic barrier for invasive fish? (International Congress of Ecology 2022 in Geneva, Switzerland). http://dx.doi.org/10.13140/RG.2.2.13227.31520 This poster won the second price of the 'best poster' competition.
Presentation	**Wiegleb, J.** (2022). Holistic management of the invasive round goby in Switzerland (ICES workshop WKSTARGATE to explore the potential for economic use of round goby and sticklebacks in the Baltic Sea in Copenhagen, Denmark)
Presentation	**Wiegleb, J.**, Hirsch, P. E., Seidel, F.& Burkhardt-Holm, P. (2020). 3D-Forces in current experienced by the round goby and two native benthic fish species during fish pass passage (Goby meeting in Starnberg, Germany)
Poster	**Wiegleb, J.**, Hirsch, P. E. & Burkhardt-Holm, P. (2018). Do round gobies overcome strong currents in fish passes? –Application of 3D-printed fish (Goby meeting in Chioggia, Italy)
Poster	**Wiegleb, J.**, Oesterwind, D., Kotterba, P. & von Nordheim, L. (2017). Trophic interactions of invasive round goby (*Neogobius melanostomus*) and Baltic herring eggs (Bonus Symposium in Thallinn) http://dx.doi.org/10.13140/RG.2.2.27651.58408
Presentation	Oesterwind, D., Bock, C., Förster, A., Gabel, M., Henseler, C., Kotterba, P., Menge, M., Myts, D., Winkler, H., **Wiegleb, J.** (2017). Integration of the invasive round goby (*Neogobius melanostomus*) into the coastal ecosystem of the Western Baltic Sea (Conference BONUS Symposium in Thallinn)

Scientific Reports

ICES. 2022. Workshop on Stickleback and Round Goby in the Baltic Sea (WKSTARGATE). ICES Scientific Reports. 4:77. 56 pp. https://doi.org/10.17895/ices.pub.21345291

Outreach activities

Energiforsk Webinar (2026): Smarter eel passage at hydroelectric power plants. Record available at Youtube: https://www.youtube.com/watch?v=za5cbPW8ikI&t=4s
Participation in the workshop "Water challenges in the mining industry" (2026), Stockholm

Wiegleb, J. (2019). Schwarzmundgrundeln in der Schweiz. *Regio Basiliensis, 60*(2), 89-96.

Presentation of the Goby Project from the Program MGU at local markets in the Kanton Basel Land in the program 'Uni am Markt'

Round goby Apéro: Stakeholder-interaction event of the round goby-research group of the Program MGU, University of Basel, Switzerland, November 2018, 2019, 2020.